

Final-year student who enjoys building real, working systems, with experience taking complex ideas from zero to deployment and thriving in environments where the problems are difficult and the pace is fast. Comfortable learning whatever is needed to move a project forward, juggling between different roles and teams when required. Multiple IEEE publications and hands-on project implementations with strong curiosity, work ethic, and a commitment to continuous learning and meaningful impact.

 Education

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| <b>Bachelor of Technology in Artificial Intelligence</b> , Amrita Vishwa Vidyapeetham, Bangalore, India  | 2022 – 2026      |
| CGPA 8.18                                                                                                                                                                                   | Bengaluru, India |
| <b>CBSE Grade 12</b> , Chrysalis High                                                                                                                                                       | 2020 – 2022      |
| Percentage: 79%                                                                                                                                                                             | Bengaluru, India |
| <b>CBSE Grade 10</b> , Sri Chaitanya Techno School                                                                                                                                          | 2019 – 2020      |
| Percentage: 92%                                                                                                                                                                             | Bengaluru, India |

 Skills

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| <b>Programming</b><br>Python, TypeScript, Java, C, SQL, HTML, CSS                                                                                                                                                                                                                                                                                                                                                                  | <b>AI &amp; Machine Learning</b> <ul style="list-style-type: none"><li>• <b>Frameworks:</b> PyTorch, TensorFlow, Ray, RLlib</li><li>• <b>Generative AI:</b> LangChain, LlamaIndex, RAG Pipelines, Transformers, Stable Diffusion APIs, LoRA</li><li>• <b>Domains:</b> Computer Vision, NLP, Time Series Forecasting, Reinforcement Learning, DRL, MARL</li></ul> |
| <b>Application Development</b> <ul style="list-style-type: none"><li>• <b>Frameworks:</b> Next.js, FastAPI, Flask</li><li>• <b>APIs:</b> REST API Design &amp; Integration, gRPC</li><li>• <b>Architecture:</b> Microservices, Scalable Systems</li></ul>                                                                                                                                                                          | <b>Cloud &amp; DevOps</b> <ul style="list-style-type: none"><li>• <b>Platforms:</b> AWS (EC2, S3, Lambda), Vercel, Heroku, Cloudflare</li><li>• <b>Containerization:</b> Docker, Minikube, Kubernetes, Grafana, Terraform</li><li>• <b>CI/CD:</b> GitHub Actions</li></ul>                                                                                       |
| <b>Core Competencies</b> <ul style="list-style-type: none"><li>• <b>Product Skills:</b> Technical Documentation, User Feedback Analysis, Market Research, Literature Review, Experimental Design, Problem-Solving &amp; Algorithmic Thinking</li><li>• <b>Soft Skills:</b> Cross-Functional Collaboration, Ownership, Results-Oriented Mindset, Team Collaboration, Project Management, Adaptability &amp; Fast Learning</li></ul> | <b>Data Engineering &amp; Processing</b> <ul style="list-style-type: none"><li>• <b>Databases:</b> PostgreSQL, MySQL, Redis, MongoDB, Pinecone, ChromaDB, ClickHouse</li><li>• <b>Data Processing:</b> Pandas, NumPy, ETL/ELT Pipelines</li><li>• <b>Data Storage:</b> AWS S3, Cloudflare R2</li></ul>                                                           |

 Projects

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| <b>High-Performance Logistics Optimization Engine</b> ,<br><i>Python, Google OR Tools, ADMM, Docker, AWS, Minikube</i> <ul style="list-style-type: none"><li>• Architected a real time DVRPTW (<b>Dynamic Vehicle Routing Problem with Time Windows</b>) solver using a hybrid OR Tools plus ADMM distributed optimization pipeline for sub second routing under volatile constraints.</li><li>• Realized 30 percent lower solve times and 20 percent higher fleet throughput via constraint decomposition and parallelized subproblem updates.</li><li>• Exposed the solver as stateless, horizontally scalable microservices with deterministic routing under live event streams.</li><li>• Deployed containerized workloads on AWS with Minikube for fault tolerant orchestration and instrumentation driven latency control.</li></ul> | 2023 – 2024 |
| <b>Multi-Agent System for Complex Operations Automation</b> ,<br><i>Python, LangGraph, ReAct Orchestration, LLM Agents, Docker, AWS, UKF, PQC</i> <ul style="list-style-type: none"><li>• Identified systemic agent drift in early prototypes and re engineered the control plane using LangGraph state graphs and deterministic delegation guarantees.</li><li>• Built a distributed multi agent reasoning stack with ReAct planning for long horizon, multi stage workflows under uncertainty.</li><li>• Integrated a custom orbital dynamics module with an Unscented Kalman Filter achieving 99.8 percent trajectory prediction accuracy and a 35 percent error reduction.</li><li>• Implemented CRYSTALS Kyber for post quantum secure channels across agent to ground communication layers.</li></ul>                                | 2024 – 2025 |

## **Adaptive Learning Platform with Real-Time Personalization**, *Next.js, FastAPI, PostgreSQL, Docker, DRL*

2024 – Present

- Designed a cognitive simulation layer using Bloom's taxonomy, spacing effects, retention decay, and frustration thresholds to formalize the learner state space.
- Benchmarked PPO, hierarchical RL, and MARL variants using controlled reward shaping and policy decomposition experiments.
- Delivered a 1.4 times increase in engagement velocity and an 18 percent retention lift through real time DRL based path adaptation.
- Raised assessment accuracy by 28 percent via continuous policy updates and telemetry driven UX refinement.
- Implemented full MLOps pipelines for automated training, safe rollout, rollback, and exploration governance at global scale.

## **Publications**

**Supply Chain Logistics with Hybrid Optimization using ADMM and Vehicle Routing Problem**, (*ICICEC*) [↗](#)

**Live Transcription and Closed Caption**, (*ICCCNT*) [↗](#)

**A Smart Pill Container for Improved Medication**, (*ICESC*) [↗](#)

**Integration of Deep Learning and Blockchain for Energy Consumption Prediction and Management in Electric Vehicles**, (*ICCCNP*)